When Does the Bootstrap Help?

A Monte Carlo Study of Confidence-Interval Coverage for the Mean under Small Samples and Skewness

Independent Computational Study

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Abstract

Bootstrap confidence intervals are often taught as a robust, assumption-light alternative to the classical Student-t interval. We test this claim with a Monte Carlo study of four interval procedures for the population mean: the Student-t interval, the bootstrap percentile interval, the bias-corrected and accelerated (BCa) interval, and the studentized bootstrap (bootstrap-t) interval. We simulate 5,000 samples for each of 20 settings formed by four populations (normal, uniform, exponential, lognormal) and five sample sizes ($n = 10$ to 160), using $B = 1{,}000$ resamples per interval. Three findings emerge. First, for symmetric populations with $n \leq 20$, the percentile and BCa intervals *undercover* (about 90% at $n = 10$ for a nominal 95%), whereas the Student-t interval is essentially exact; we show that a simple "narrowness" argument predicts this shortfall to within half a percentage point. Second, under strong right skewness all four methods undercover at small n, but the bootstrap-t interval is clearly the most accurate (90.8% versus 80–84% for the others in the lognormal case at $n = 10$). Third, the errors of the non-studentized methods are highly asymmetric: almost all misses occur because the interval falls entirely below the true mean. The price of the bootstrap-t interval's accuracy is width, which can exceed twice that of the t interval in small, heavily skewed samples. We conclude with practical recommendations for choosing an interval in applied work.

Keywords: bootstrap; confidence intervals; coverage probability; BCa; studentized bootstrap; Monte Carlo simulation; skewness

1 Introduction

Since its introduction by Efron (1979), the bootstrap has become one of the most widely used tools in applied statistics. Its appeal is easy to state: rather than relying on a formula derived under a parametric model, we approximate the sampling distribution of a statistic by resampling the observed data (Efron and Tibshirani, 1993). In introductory courses the bootstrap is frequently presented as the method to reach for when the data "do not look normal".

This framing hides an important subtlety. The theoretical guarantees of the bootstrap are *asymptotic*. In small samples, some bootstrap intervals can be noticeably less accurate than the classical Student-t interval (Student, 1908) that they are meant to improve upon (Hesterberg, 2015). At the same time, more refined bootstrap intervals, namely the BCa interval (Efron, 1987) and the bootstrap-t interval, are known to have better higher-order accuracy than both the percentile and the normal-theory intervals (Hall, 1992; DiCiccio and Efron, 1996). Which effect dominates at the sample sizes typical of a laboratory experiment or a class project (n between 10 and 50) is ultimately an empirical question.

This paper answers that question for the simplest possible target, the population mean, using a controlled Monte Carlo experiment. Our contributions are:

1. A side-by-side comparison of four widely used intervals over 20 population–sample-size combinations, with Monte Carlo error small enough (≈ 0.3 percentage points) to separate the methods reliably.
2. A decomposition of non-coverage into its two tails, showing *how* each method fails under skewness, not just how often.
3. A short analytic explanation of the percentile interval's small-sample undercoverage under normality, whose predictions match the simulation closely.
4. Practical guidance on when each interval is appropriate.

2 Interval Procedures

Let $X_1, \ldots, X_n$ be an i.i.d. sample with mean μ, and let $\bar{X}$ and s denote the sample mean and standard deviation, with $\widehat{\mathrm{se}} = s/\sqrt{n}$. Let $\bar{X}_1^*, \ldots, \bar{X}_B^*$ be the means of B bootstrap resamples drawn with replacement from the data. All intervals below target 95% coverage ($\alpha = 0.05$).

Student-t. The classical interval is $\bar{X} \pm t_{n-1,\,1-\alpha/2}\,\widehat{\mathrm{se}}$. It is exact for normal data and is the baseline against which we compare.

Percentile bootstrap. The interval is $[\hat{q}_{\alpha/2},\, \hat{q}_{1-\alpha/2}]$, where $\hat{q}_p$ is the p-quantile of the bootstrap means. It requires no variance formula and automatically reflects some skewness in the bootstrap distribution.

BCa bootstrap. The BCa interval (Efron, 1987) uses adjusted percentiles

$$\alpha_j = \Phi\left(\hat{z}_0 + \frac{\hat{z}_0 + z_j}{1 - \hat{a}(\hat{z}_0 + z_j)}\right), \qquad j \in \{\alpha/2,\ 1-\alpha/2\}, \tag{1}$$

where $z_j = \Phi^{-1}(j)$, the bias correction is $\hat{z}_0 = \Phi^{-1}\big(\#\{\bar{X}_b^* < \bar{X}\}/B\big)$, and the acceleration $\hat{a}$ is estimated by the jackknife:

$$\hat{a} = \frac{\sum_i (\bar{\theta}_{(\cdot)} - \hat{\theta}_{(i)})^3}{6\left[\sum_i (\bar{\theta}_{(\cdot)} - \hat{\theta}_{(i)})^2\right]^{3/2}}, \tag{2}$$

with $\hat{\theta}_{(i)}$ the mean of the sample with observation i removed and $\bar{\theta}_{(\cdot)}$ their average. The interval is $[\hat{q}_{\alpha_{\mathrm{lo}}},\, \hat{q}_{\alpha_{\mathrm{hi}}}]$.

Bootstrap-t (studentized). For each resample we compute $T_b^* = (\bar{X}_b^* - \bar{X})/\widehat{\mathrm{se}}_b^*$, where $\widehat{\mathrm{se}}_b^*$ is the standard error computed from resample b. With $\hat{t}_p$ the p-quantile of the T_b^*, the interval is

$$\left[\bar{X} - \hat{t}_{1-\alpha/2}\,\widehat{\mathrm{se}},\ \ \bar{X} - \hat{t}_{\alpha/2}\,\widehat{\mathrm{se}}\right]. \tag{3}$$

Note the reversal of quantiles: if the bootstrap t-distribution has a long left tail (as it does for right-skewed data), the interval is stretched to the *right*.

Known theory. For equal-tailed two-sided intervals, the leading $O(n^{-1/2})$ errors in the two tails cancel, so all four methods have two-sided coverage error of order $O(n^{-1})$. The methods differ in their *one-sided* accuracy: the Student-t and percentile intervals have one-sided errors of order $O(n^{-1/2})$, while BCa and bootstrap-t are second-order accurate, with one-sided errors of order $O(n^{-1})$ (Hall, 1992; DiCiccio and Efron, 1996). Asymptotic orders, however, say nothing about the size of the constants at small n, which motivates the simulation.

3 Simulation Design

Populations. We use four populations chosen to span symmetric and skewed shapes: Normal$(0,1)$ (skewness 0), Uniform$(0,1)$ (skewness 0, light tails), Exponential(1) (skewness 2), and Lognormal$(0,1)$ (skewness ≈ 6.18, heavy right tail), whose mean is $e^{1/2} \approx 1.649$.

Sample sizes and replication. We use $n \in \{10, 20, 40, 80, 160\}$, giving 20 settings. For each setting we draw $R = 5{,}000$ independent samples and compute all four intervals on each, using the same $B = 1{,}000$ bootstrap resamples for the three resampling methods (common random numbers), which sharpens comparisons between methods.

Metrics. For each method we record (i) *coverage*, the proportion of intervals containing μ; (ii) the *lower* and *upper non-coverage* rates, i.e., the proportions with $\mu < L$ and $\mu > U$ respectively, each ideally equal to 0.025; and (iii) the mean interval width. With $R = 5{,}000$, the Monte Carlo standard error of a coverage estimate near 0.95 is $\sqrt{0.95 \times 0.05/5000} \approx 0.0031$. We therefore treat estimates in $[0.944, 0.956]$ (nominal ± 2 standard errors) as consistent with exact coverage.

Implementation. All computations were carried out in Python with NumPy and SciPy, using a single fixed random seed for reproducibility. Bootstrap quantiles use linear interpolation; the BCa bias-correction proportion is clipped to $[1/(2B),\ 1 - 1/(2B)]$ to avoid infinite $\hat{z}_0$.

4 Results

Figure 1 shows coverage as a function of n for each population, and Table 1 reports the full set of estimates.

4.1 Symmetric populations: the t interval wins

For normal data, the Student-t interval achieves nominal coverage at every sample size, as theory guarantees. The bootstrap-t interval performs equally well. In contrast, the percentile and BCa intervals undercover noticeably at small n: at $n = 10$ their coverage is 0.900 and 0.898, i.e., the actual error rate is roughly double the nominal 5%. The shortfall shrinks steadily and is negligible by $n = 160$.

The uniform population tells a similar story with one twist. The percentile interval again undercovers at small n (0.919 at $n = 10$); BCa does better here (0.937). The bootstrap-t interval, however, *overcovers* (0.976 at $n = 10$), meaning its intervals are wider than necessary. For light-tailed data the sample standard deviation is relatively stable, so studentizing each resample adds variability without correcting any real skewness.

4.2 Skewed populations: the bootstrap-t interval wins

Skewness changes the ranking. For the exponential population at $n = 10$, the coverage of the Student-t, percentile, BCa, and bootstrap-t intervals is 0.900, 0.862, 0.875, and 0.942. Only the bootstrap-t interval comes close to nominal. By $n = 40$, BCa and Student-t reach about 0.93, and by $n = 160$ all methods are within about one percentage point of 0.95.

The lognormal population is the hardest case. Even the best method, bootstrap-t, achieves only 0.908 at $n = 10$, while the others range from 0.804 (percentile) to 0.839 (Student-t). Convergence is slow: at $n = 160$, the percentile and Student-t intervals still cover only about 92%. Notably, *BCa never improves on the classical t interval by more than one percentage point* in any of our 20 settings, despite its better

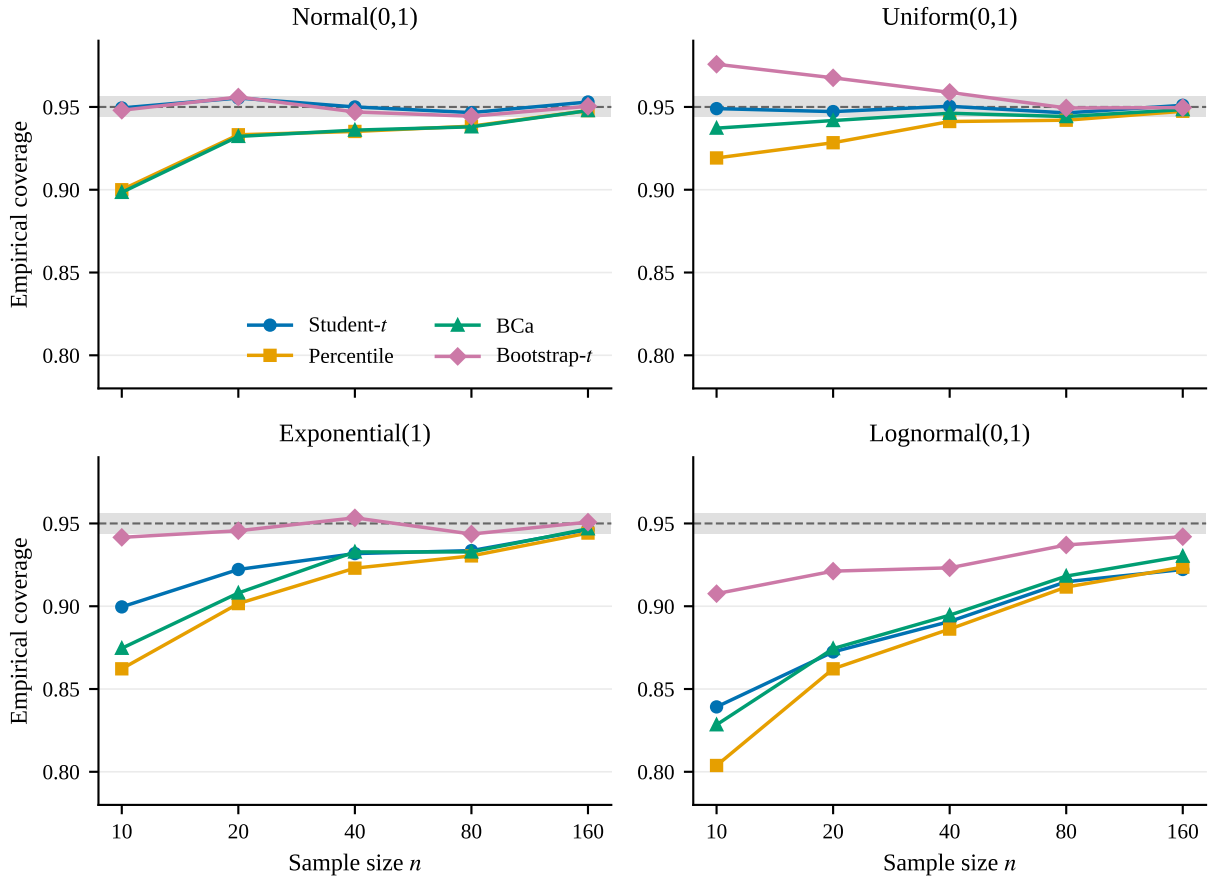


Figure 1: Empirical coverage of nominal 95% intervals for the mean ($R = 5{,}000$ samples per point). The dashed line marks 0.95; the grey band is ± 2 Monte Carlo standard errors.

asymptotic properties, and it is substantially worse at small n for symmetric and moderately skewed data. Its advantage over the percentile interval is consistent but small.

4.3 How the intervals fail: tail asymmetry

Total coverage hides an important pattern. Figure 2 splits non-coverage into its two tails for the skewed populations at $n = 20$. For an ideal equal-tailed interval each bar would be at 0.025.

For both skewed populations, nearly all failures are *upper misses*: the interval lies entirely below the true mean. For the lognormal population, the Student-t interval misses below μ in 12.7% of samples while essentially never missing above it (0.0%). The mechanism is intuitive: with a long right tail, small samples often contain none of the rare large values, so both $\bar{X}$ and s are too small *together*, and a symmetric interval centered at $\bar{X}$ cannot reach μ. The percentile interval inherits the same problem because the bootstrap distribution of $\bar{X}^*$ is built only from the values that happened to be observed.

BCa partially rebalances the tails (lognormal: 2.4% lower, 10.2% upper), and the bootstrap-t interval does so most effectively (1.0% and 6.9%). This is exactly the one-sided, second-order improvement predicted by theory: the bootstrap-t interval stretches toward the long tail because samples with small $\bar{X}$ also tend to have small $\widehat{\mathrm{se}}^*$, producing large negative values of T^*.

Table 1: Empirical coverage of nominal 95% intervals. Bold entries lie within ± 2 Monte Carlo standard errors of 0.95 (i.e., in $[0.944, 0.956]$).

Population	n	Student-t	Percentile	BCa	Bootstrap-t
Normal(0,1)	10	**0.949**	0.900	0.898	**0.948**
	20	**0.955**	0.933	0.932	**0.956**
	40	**0.950**	0.935	0.936	**0.947**
	80	**0.947**	0.938	0.938	**0.944**
	160	**0.953**	**0.948**	**0.948**	**0.950**
Uniform(0,1)	10	**0.949**	0.919	0.937	0.976
	20	**0.947**	0.928	0.942	0.968
	40	**0.950**	0.941	**0.946**	0.959
	80	**0.946**	0.942	**0.944**	**0.949**
	160	**0.951**	**0.947**	**0.948**	**0.950**
Exponential(1)	10	0.900	0.862	0.875	0.942
	20	0.922	0.902	0.908	**0.946**
	40	0.932	0.923	0.933	**0.953**
	80	0.934	0.930	0.933	0.944
	160	**0.946**	**0.944**	**0.947**	**0.951**
Lognormal(0,1)	10	0.839	0.804	0.828	0.908
	20	0.872	0.862	0.874	0.921
	40	0.891	0.886	0.895	0.923
	80	0.915	0.912	0.918	0.937
	160	0.922	0.924	0.930	0.942

4.4 The cost of accuracy: interval width

Table 2 reports the mean width of each bootstrap interval relative to the Student-t interval for the skewed populations.

The percentile interval is the narrowest in every small-sample setting, which explains much of its undercoverage. The bootstrap-t interval is the widest: at $n = 10$ in the lognormal case it is on average 2.13 times as wide as the t interval, and its width varies considerably from sample to sample. This is the familiar trade-off: honest coverage under heavy skewness requires admitting greater uncertainty. As n grows, all ratios approach one.

5 Discussion

5.1 Why the percentile interval is too narrow

The percentile interval's undercoverage under *normality* can be explained almost exactly. For the sample mean, the bootstrap distribution of $\bar{X}^*$ is approximately normal with standard deviation $\hat{\sigma}/\sqrt{n}$, where $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$ uses divisor n rather than $n-1$. Its quantiles are therefore approximately $\bar{X} \pm z_{0.975}\sqrt{(n-1)/n}\,\widehat{\mathrm{se}}$. Compared with the t interval, the percentile interval is shrunk by the factor

$$\rho_n = \sqrt{\frac{n-1}{n}} \cdot \frac{z_{0.975}}{t_{n-1,\,0.975}}. \tag{4}$$

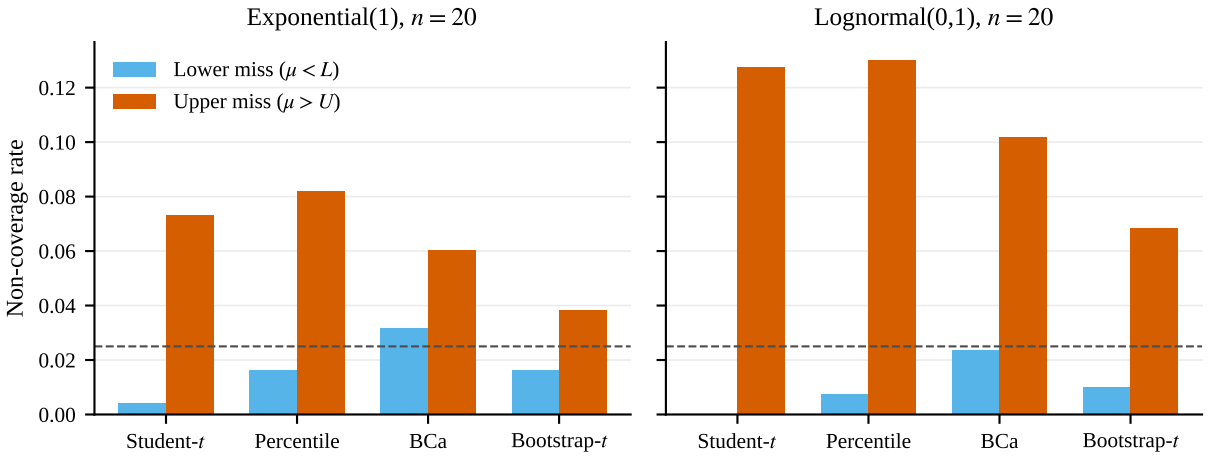


Figure 2: Non-coverage split by direction at $n = 20$. "Upper miss" means the whole interval lies below the true mean. The dashed line marks the ideal rate of 0.025 per tail.

Table 2: Mean width of the Student-t interval and width of each bootstrap interval relative to it (ratio > 1 means wider).

Population	n	Student-t width	Percentile	BCa	Bootstrap-t
Exponential(1)	10	1.327	0.81	0.87	1.39
	20	0.893	0.90	0.96	1.17
	40	0.625	0.95	0.99	1.08
	80	0.440	0.97	0.99	1.04
	160	0.310	0.98	1.00	1.02
Lognormal(0,1)	10	2.444	0.79	0.91	2.13
	20	1.739	0.89	1.02	1.56
	40	1.251	0.94	1.05	1.32
	80	0.903	0.97	1.05	1.19
	160	0.646	0.98	1.03	1.11

Two separate sources of narrowness appear: the n divisor in the variance, and the use of normal rather than t quantiles, which ignores the uncertainty in estimating σ. For normal data, the implied coverage is $P\big(|T_{n-1}| \le z_{0.975}\sqrt{(n-1)/n}\big)$ with $T_{n-1} \sim t_{n-1}$. At $n = 10$ this gives $\rho_{10} = 0.822$ and a predicted coverage of 0.904; the simulation gives a width ratio of 0.814 and a coverage of 0.900. At $n = 20$ the predictions are 0.913 and 0.929, versus observed values of 0.907 and 0.933. Since BCa is built on the same bootstrap distribution, it inherits this narrowness when there is little skewness to correct.

5.2 Practical recommendations

Our results suggest the following guidance for inference on a mean:

- **Roughly symmetric data, any n:** use the Student-t interval. The percentile and BCa intervals offer no benefit and undercover when $n \lesssim 20$.
- **Moderately or strongly right-skewed data, small n:** prefer the bootstrap-t interval, accepting that it will be wider. Report the interval honestly rather than switching methods to obtain a narrower one.

- **Avoid the plain percentile interval for small samples.** It had the lowest coverage in 16 of the 20 settings, and in each of the remaining four it was within 0.2 percentage points of the lowest.
- **Very heavy tails and $n < 40$:** no method reaches nominal coverage. Consider whether the mean is the right target (a median or a mean on a log scale may be more stable), or collect more data.

5.3 Limitations

This study has several limitations. We considered only the mean; for statistics such as the median or a correlation coefficient, no simple t-type formula exists and the relative merits of bootstrap methods may differ (Davison and Hinkley, 1997). We studied four populations, and in particular no left-skewed or multimodal distributions, although by symmetry left skewness would simply mirror our results. We used $B = 1{,}000$ resamples, which is adequate for 95% intervals but adds a small amount of extra variability to the endpoints. Finally, the bootstrap-t interval relies on a stable standard-error estimate within each resample; for statistics without a simple standard-error formula, a nested bootstrap is required, which is considerably more expensive.

6 Conclusion

The bootstrap is not automatically more robust than classical methods. For symmetric populations and small samples, the simple Student-t interval is more accurate than the percentile and BCa bootstrap intervals, and a short calculation explains why. For skewed populations, the studentized bootstrap-t interval offers a real and substantial improvement, mainly by correcting the strong tendency of other intervals to fall below the true mean. Both observations are consistent with established asymptotic theory, but the simulation shows that the differences are large enough to matter at the sample sizes common in practice. The choice of interval should therefore be guided by the shape of the data and the sample size, not by a general preference for resampling.

Reproducibility and disclosure. All results were produced by a single Python script (NumPy 2.5, SciPy 1.18) with a fixed random seed; the full simulation runs in about two minutes on a laptop-class CPU. The simulation code and text of this report were prepared with the assistance of an AI system (Claude, Anthropic); all reported numbers come directly from the executed code.

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